Checking the Blockmodel
A Bayesian Workflow for the Literatures Step

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Abstract. The citation blockmodel returned a consensus partition of 1,268 groups for 3,557 research communities. Nearly half the communities lacked a stable assignment across posterior draws, and two runs from the same random seed returned different hierarchies. When we scored cuts of the fitted hierarchy with a flat degree-corrected blockmodel, the 405-group cut had the shortest description among the cuts examined. Before using any cut to estimate how the effect of topical proximity varies across literatures, we need to determine how the treatment of citation counts, the hierarchy prior, and the fitting procedure each contribute to the result. The description-length audit has been run; everything that follows it here has not. What follows is the plan: three stages of checking – prior pushforward, prior predictive, and posterior – with what each measures, what domain knowledge it needs, and what each outcome would oblige us to do.

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1 The decision this document serves

Model I – the community-pair model of §13.2 of the specification outline, reproduced as equation (1) below – cannot be estimated until every research community carries a label saying which literature it belongs to. A blockmodel was fitted to supply those labels. It returned a hierarchy thirteen levels deep whose base divides 3,557 communities into 1,268 groups, left 1,771 of those communities without a stable assignment across posterior draws, and produced different answers on two runs from the same random seed.

Scoring cuts of that hierarchy under a flat degree-corrected blockmodel gives a curve with one minimum, at the 405-group cut, and one near-free merge, into 168 groups. Every coarser cut adds between 19 and 33 percent of the shortest description found. The base of the hierarchy – the 1,268 groups – scores 80 percent longer than the 405-group cut. (The consensus puts 1,268 groups at the base; the audit, which works from the posterior draws, scores a median of 1,256. §4 reconciles the counts.)

Three candidate explanations remain open, and the workflow in §7 to §9 exists to separate them: the treatment of citation counts, which fixes how much the model believes it has seen; the hierarchy prior, which fixes how cheaply fine partitions can be described; and the fitting procedure itself, whose non-reproducibility is unexplained. The decision that waits on all three is which cut of the hierarchy becomes the literature label ℓ in Model I.

2 What a literature is, and why the model needs one

Model I asks a question about pairs of research communities: when two communities work on similar questions, do they draw on the same earlier literature more often than two communities that do not? Shared citation is the observable, topical proximity is the predictor, and the coefficient linking them is the quantity of interest.

That coefficient is not allowed to be a single number. It varies two ways. It drifts across the six era pairs, which asks whether the field's tendency to connect similar work has changed over fifty years. And it varies across *literatures*, which asks whether the tendency works the same way everywhere in the field or differently in different parts of it. Writing S_{ijp} for the topical proximity of communities i and j in era pair p,

$$\log \lambda_{ijp} = \alpha_p^K + \log E_{ijp}^K + \underbrace{\alpha_{S,p}\, S_{ijp}}_{\text{drifts over time}} + \underbrace{\alpha_{S,\ell}\, S_{ijp}}_{\text{varies by literature}} + \boldsymbol{z}_{ijp}^{\top}\boldsymbol{\gamma} + \theta_i^K + \psi_j^K. \tag{1}$$

The superscript K marks the coupling model throughout, distinguishing its era baselines, exposures, and community effects from those of the citation model that shares the specification. θ_i^K and ψ_j^K are random intercepts for the citing and cited communities.

The second of those terms needs an ℓ: a label on every community saying which literature it belongs to. Nothing else in the project supplies one. Journals do not, because a cognitive paper can appear in a pharmacology journal. Keywords do not, because they describe subject matter rather than position. Communities themselves do not, because they are found separately within each seven-year era and carry no identity across eras.

A literature, in the sense the model needs, is a group of communities that connect to the rest of the field in the same way. Two pharmacokinetics communities in adjacent eras may barely cite one another and still belong together, because the rest of the field draws on them interchangeably. That

criterion – sameness of position rather than density of contact – is what a blockmodel estimates, and it is why a blockmodel is in the pipeline at all.

The specific instrument is a *nested degree-corrected stochastic blockmodel*, fitted with the `graph-tool` library. Three words in that name do real work. *Stochastic blockmodel*: communities are sorted into groups so that the citation traffic between any two groups is described by a single rate, and the sorting that describes the traffic most economically wins. *Degree-corrected*: large communities have more connections by virtue of being large, and the model accounts for that so the groups reflect affinity rather than volume. *Nested*: the groups are themselves grouped, and those groups grouped again, so the result is a hierarchy rather than a single partition.

3 What the fit returned

The network handed to the blockmodel has one node per community – 3,557 of them, across eras 11 to 17 – and a directed edge wherever one community's works cite another's, weighted by the number of citations. It holds 36,105 distinct community pairs and 583,955 citations, with pair weights running from 1 to 4,646.

Table 1 records what came back. Each row is a result that would be unremarkable on its own and is not unremarkable in company.

4 The audit that turned a suspicion into a measurement

Three group counts appear in this document and they are three different objects. The *point estimate* from the description-length heuristic has 846 groups at its base. The *consensus*, formed by reconciling 250 of the 499 posterior draws, has 1,268. The *median across the 50 draws* scored in the audit below has 1,256. Nothing is inconsistent here; a consensus of disagreeing partitions tends toward their intersection, which is why it is finer than the estimate it summarizes.

A hierarchy offers a choice of granularity, and the obvious question is which level to treat as the literature. The hierarchy itself cannot answer: it runs 1,268, 405, 168, 82, 44, 25, 14, 8, 4, 2, 1, with successive ratios of 3.1, 2.4, 2.0, 1.9, and then 1.8 to 2.0 the rest of the way down. The base thins fastest and the lower levels are steady; nowhere is there a plateau or a gap. A dendrogram drawn from it is no help either, because its horizontal axis is level number rather than any measure of dissimilarity – the branches have no lengths, so nothing in the picture marks a natural cut.

Description length supplies the lengths. Take each level's grouping, project it back onto the 3,557 communities, hold it fixed, and ask what a flat degree-corrected blockmodel with that grouping costs to describe. Coarsening always loses fit and always saves complexity; the level where the loss begins to outrun the saving is the granularity the network supports.

Two readings follow, and they point in opposite directions. The base of the hierarchy scores *80 percent longer* under the flat model than the 405-group cut does. Whatever produced 1,268 groups, it was not a flat accounting of the citation record. But coarsening scores badly too: every merge below 168 groups adds between 19 and 33 percent of the shortest description found, so a reading of eight literatures gives up more than the whole of that description.

Two cautions on reading Table 2. The comparison is among *cuts of the fitted hierarchy*, scored under a flat model; no flat partition was optimized independently, so 405 is the lowest-scoring cut examined rather than the best flat partition available. And the scoring model is not the model that produced the hierarchy – the nested fit's own description length is about 225,000 nats, far below anything in the table, because the hierarchical prior of §5 charges much less for the same

Table 1: What the fit returned, and why each result is a problem

Result	Value	Why it is a problem
Groups at the base of the hierarchy	1,268	3,557 communities in 1,268 groups is 2.8 communities apiece. Most groups would then contribute a handful of pairs over a narrow range of topical proximity, leaving σ_ℓ estimated largely from its prior. See §11.
Groups holding one community	690 (54%)	Half the groups are not groups. The communities in them are not marginal either: their median degree is 34.5 against 5 for the rest, and their median size is 89 works against 4.
Communities with no settled assignment	1,771 (49.8%)	Across posterior draws, half the communities fail to place in any one group on 90 percent of draws. The consensus partition is an average of disagreeing answers.
Consensus against point estimate	1,268 vs 846	The consensus is *finer* than the fit it summarizes. Averaging partitions that disagree produces their intersection, and intersections splinter.
Reproducibility	two answers	Two runs on identical input with the same seed returned different hierarchies, 2,451 nats – natural-log units of information – apart in description length.
Compute	15–20 hours	Of which 7.6 hours was reconciling draws, a step whose cost scales with the number of groups being matched. 499 were collected, every second one aligned, so 250 entered the consensus.

granularity. The table prices cuts on a common scale; it does not adjudicate between the two models.

The fifty posterior draws lie almost on top of one another in this curve. The shape is not one fit's idiosyncrasy.

The audit also recorded three facts about particular cuts that description length does not capture, and §11 leans on them. At the 25-group cut, assignment confidence – the share of draws placing a community in its modal group – stops improving as the cut coarsens; 24 of the 25 groups span more than one era; and no group is a singleton. At the 8-group cut, two of the eight groups hold 2 percent of the communities between them, so a reading of eight literatures is in practice a reading of six.

Table 2: Flat-model description length of each cut of the fitted hierarchy, median over 50 posterior draws. The three degenerate top levels, all of one group, are omitted.

Groups	DL (nats)	Merge cost	vs best	Reading
1,256	1,327,232	—	+80.1%	base of the hierarchy
405	736,908	−590,324	—	lowest-scoring cut examined
168	750,206	+13,298	+1.8%	the only near-free merge
82	971,404	+221,198	+31.8%	
44	1,120,231	+148,827	+52.0%	
25	1,256,995	+136,764	+70.6%	assignment confidence stops improving; no singletons
14	1,463,906	+206,911	+98.7%	
8	1,646,272	+182,366	+123.4%	two groups hold 2 percent between them
4	1,868,082	+221,810	+153.5%	
2	2,110,950	+242,868	+186.5%	

5 The model being fitted

The assumptions at issue live in particular terms of this model, so it is set out here rather than named. The account is Peixoto's, as implemented in `graph-tool`; the notation is his.

5.1 One partition

Write $N = 3{,}557$ for the number of communities and $\boldsymbol{A}$ for the matrix of citation counts, so that A_{ij} is the number of citations from community i's works to community j's. A partition $\boldsymbol{b}$ assigns every community a group, $b_i \in \{1, \dots, B\}$. Two quantities summarize what the partition sees: the citations running between groups,

$$e_{rs} \;=\; \sum_{ij} A_{ij}\, \delta_{b_i r}\, \delta_{b_j s},$$

with margins $e_r^{\text{out}} = \sum_s e_{rs}$ and $e_s^{\text{in}} = \sum_r e_{rs}$; and the degrees k_i^{out} and k_i^{in} of each community.

The generative story is that citations arrive as independent Poisson draws whose rate depends on the groups at either end and on how active each community is:

$$A_{ij} \sim \text{Poisson}\big(\theta_i^{\text{out}}\, \theta_j^{\text{in}}\, \lambda_{b_i b_j}\big)\,. \tag{2}$$

Here λ_{rs} is the affinity between groups r and s, and θ_i^{out}, θ_i^{in} are the propensities of the individual community. Those θ terms are the degree correction. Without them a community that cites heavily would be grouped with other heavy citers whatever it cites, and with community sizes in this corpus running from 2 works to 2,769, that failure mode is not hypothetical.

Maximizing over $\boldsymbol{\theta}$ and $\boldsymbol{\lambda}$ leaves a quantity that depends only on the partition,

$$\mathcal{S}(\boldsymbol{b}) \;=\; -\sum_{rs} e_{rs}\, \ln \frac{e_{rs}}{e_r^{\text{out}} e_s^{\text{in}}}, \tag{3}$$

which is how many nats – units of information, like bits but measured in natural logarithms – it takes to transmit the citation matrix once the receiver knows the partition. Smaller is better, and $\mathcal{S}$ falls monotonically as groups are added – a partition with one community per group describes the data perfectly and explains nothing.

5.2 What stops it from using every group it can

The brake is the cost of transmitting the model itself. The full description length adds to (3) the cost of the partition, the between-group counts, and the degrees:

$$\Sigma \;=\; \underbrace{\mathcal{S}(\boldsymbol{b})}_{\text{the data}} \;+\; \underbrace{\mathcal{L}}_{\text{the model}}, \qquad \mathcal{L} = -\ln P(\boldsymbol{e}) - \ln P(\boldsymbol{k} \mid \boldsymbol{e}, \boldsymbol{b}) - \ln P(\boldsymbol{b}). \tag{4}$$

Minimizing Σ is identical to maximizing the posterior $P(\boldsymbol{b} \mid \boldsymbol{A})$ under those priors. The three terms of $\mathcal{L}$ *are* the priors, and they are these:

$$P(\boldsymbol{b}) = \frac{\prod_r n_r!}{N!} \binom{N-1}{B-1}^{-1} \frac{1}{N}, \qquad \text{group sizes } n_r\text{, uniform given } B, \tag{5}$$

$$P(\boldsymbol{e}) = \left(\!\!\binom{B^2}{E}\!\!\right)^{-1}, \qquad \text{uniform over count matrices,} \tag{6}$$

$$P(\boldsymbol{k} \mid \boldsymbol{e}, \boldsymbol{b}) = \prod_r \left(\!\!\binom{n_r}{e_r^{\text{out}}}\!\!\right)^{-1} \left(\!\!\binom{n_r}{e_r^{\text{in}}}\!\!\right)^{-1}, \qquad \text{uniform over degree sequences,} \tag{7}$$

where $\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}$ counts the ways to choose k items from n types with repetition, and E is the total number of edges. The directed model carries separate in- and out-degree terms, as shown.

One caution on equation (7). It is the *uniform* degree prior, which `graph-tool` offers as an option; the library's default is `degree_dl_kind="distributed"`, a non-uniform hyperprior that models the degree distribution itself. Which was in force during our fit has not been checked, and it should be, because a hyperprior on degrees is one of the few places in this model where something resembling domain knowledge could enter.

What the three terms have in common is that each is uniform over some combinatorial set, and none of them mentions citation, or science, or anything else about where the data came from.

5.3 From one partition to a hierarchy

The flat prior $P(\boldsymbol{e})$ above has a known and unwanted consequence: it cannot resolve more than about $\sqrt{E}$ groups, because the cost of describing a $B \times B$ count matrix grows faster than the fit improves. The ceiling is a function of how many edges the model believes it has, which is why §5.4 matters here too: at the 36,105 distinct pairs it is on the order of 190, and at the $E = 583{,}955$ the multiplicity treatment declares it is on the order of 760. The constant in front of $\sqrt{E}$ depends on the average degree, so these are magnitudes rather than values. Peixoto states the limit in the familiar form $O(\sqrt{N})$, which coincides with $\sqrt{E}$ only for sparse graphs with $E = O(N)$; under the multiplicity treatment ours is not sparse in that sense, and $\sqrt{N} = 60$ would be off by an order of magnitude. Either way it is a detection limit – a statement about what the flat model can resolve, not about how many literatures the field contains.

The nested model removes the ceiling by refusing to describe the count matrix flatly. Instead it treats $\boldsymbol{e}$ as a network in its own right – one with B_1 nodes – and partitions *that*, recursively, until one group remains:

$$P(\{\boldsymbol{e}_l\}, \{\boldsymbol{b}_l\}) \;=\; \prod_{l=1}^{L} P(\boldsymbol{e}_l \mid \boldsymbol{e}_{l+1}, \boldsymbol{b}_l)\, P(\boldsymbol{b}_l), \qquad B_L = 1. \tag{8}$$

Our fit has $L = 13$ levels, with $B_1 = 1{,}268$ at the base. The top three collapse to a single group and carry no information, which is why Table 2 lists ten cuts rather than thirteen.

Equation (8) is the single most consequential line in this document. It is the reason a grouping into 1,268 blocks is cheap to transmit: the blocks need not be described one by one against a flat background, only against the coarser grouping above them. Note where this assumption sits. It is a prior over the count matrices, not over the partition – the marginal prior on the number of groups is uniform, as §7 shows – so it reaches $\boldsymbol{b}$ through the marginal likelihood. That is why the instrument for probing it is the prior predictive and not a pushforward. Equation (8) also fixes the prior over depth: each B_l is drawn uniformly from $\{1, \ldots, B_{l-1}\}$, so the hierarchy is a chain that descends from N to one group in a random number of steps.

5.4 Weights, and the quantity E

Citation counts enter through E, and E appears in $P(\boldsymbol{e})$. Passing the counts as `eweight` declares them to be edge multiplicities: a pair joined by 4,646 citations contributes 4,646 to E, so $E = 583{,}955$ rather than the 36,105 distinct community pairs. Since $P(\boldsymbol{e})$ is a multiset coefficient over B^2 cells and E draws, a larger E supports a larger B at the same cost. The likelihood choice and the prior are therefore coupled through a single integer, which is why §10 settles the weights before checking anything else.

5.5 How it was fitted, and with what

The implementation is `graph-tool` 3.9, Tiago Peixoto's C++ library with Python bindings, installed in its own `conda` environment at `/opt/gt-env` and invoked by absolute path. The version string and the argument names below were read from the installed library rather than from documentation: an earlier script passed `state_args`, the name used in the 2.x series, and 3.9 warned of `unrecognized keyword arguments` and discarded it. Every call here is written against signatures confirmed by inspection. Four calls do the work:

- `minimize_nested_blockmodel_dl` – an agglomerative heuristic that searches for the partition minimizing (4), returning a point estimate in about 45 seconds. Degree correction and the weights enter here, through `base_state_args`.
- `mcmc_equilibrate`, with merge-and-split proposals – Markov chain Monte Carlo over partitions. Starting from the heuristic's answer it found a description length 18,329 nats lower, so the heuristic alone was well short of the mode.
- `PartitionModeState` – group labels are arbitrary and change between draws, so nothing can be averaged until the draws are put on a common labeling. Of the 499 draws collected, every second one was aligned; those 250 took 7.6 hours.
- `get_max_nested` and `get_marginal` – the consensus partition and, for each community, the share of draws placing it in each group.

`graph-tool` exposes arguments for the likelihood – degree correction on or off, weights as multiplicities or as covariates, nested or flat. It exposes no argument for $P(\boldsymbol{b})$. The priors in equations (5) to (7) are the ones we get.

6 Where the assumptions are

Two choices shaped everything above, and neither was examined before the fit.

6.1 The likelihood: citation counts as multiplicities

Section 5.4 gave the mechanism; what matters here is that it was never examined. `eweight` declares the counts to be edge multiplicities, so the pair joined by 4,646 citations enters the likelihood as 4,646 parallel edges rather than as one relationship of high intensity. That single argument fixes E, E fixes $P(\boldsymbol{e})$, and $P(\boldsymbol{e})$ is one of the three terms that decides how many groups the model will tolerate.

The alternative treats the count as a covariate attached to one edge, through `recs` with a discrete Poisson record type. That is a different generative story about the same observations, and the one usually preferred when weights are heavily skewed. Neither is obviously right. What is wrong is having chosen between them in a single unremarked line.

6.2 The prior: reference priors with no domain content

`graph-tool`'s priors over partitions are uniform within microcanonical ensembles – every arrangement consistent with a fixed set of totals gets equal weight – chosen so that the description length is a clean coding argument with nothing tuned. They encode nothing whatever about citation networks. Nothing in them says that in-degree is heavy-tailed, that reference lists are bounded, that citation runs backwards in time and almost never forwards, or that a field of this size holds tens of specialties rather than thousands.

Being empty of domain content does not make them weak in effect. The nested prior exists precisely to make deep hierarchies with many small blocks cheap to describe, and Table 2 measures how much that costs once the subsidy is withdrawn. We therefore have a prior with no knowledge in it and large consequences out of it: not uninformative where it matters, and not informed where it should be.

There is a domain-informed prior available, and it has been stated in conversation without being written down: a field of this size, over fifty years, holds something like twenty to thirty literatures. That is a weakly informative claim, defensible from the co-citation literature on specialty structure, and the model had no access to it. Formalizing it is the business of §7.

6.3 Why the order matters

The two cannot be checked independently. Informativeness is a property of the prior and the likelihood together, not of the prior alone – a flat prior on a logistic regression coefficient is wildly informative about outcomes. Here, the same prior over partitions implies one predictive distribution when the model believes it has 583,955 observations and quite another when it believes it has 36,105. Settling the likelihood therefore comes first, and every check below assumes it has been settled.

7 Stage one: prior pushforward checks

A pushforward check pushes the prior through a *function of the parameters* and inspects the distribution that comes out. It stops short of generating data, which makes it cheap, and it is the natural first instrument whenever the quantity in dispute is itself a parameter.

The quantity in dispute is the number of groups. Nobody has intuitions about a prior over partitions of 3,557 objects; everybody has intuitions about how many literatures a field contains. The pushforward is supposed to convert the first into the second.

7.1 What the prior over B actually is

It can be read off equation (5) without sampling anything. The partition prior factors as $P(\boldsymbol{b}) = P(\boldsymbol{b} \mid \boldsymbol{n})\, P(\boldsymbol{n} \mid B)\, P(B)$ with $P(B) = 1/N$, and the remaining two terms of $\mathcal{L}$ are normalized conditionals – $P(\boldsymbol{e})$ given B and E, and $P(\boldsymbol{k})$ given $\boldsymbol{e}$ and $\boldsymbol{b}$ – so summing them out leaves $P(\boldsymbol{b})$ unchanged. The marginal prior over the number of groups is therefore **uniform on** $\{1, \ldots, 3{,}557\}$, with mean $1{,}779$. The same holds at every level: B_l is uniform on $\{1, \ldots, B_{l-1}\}$.

Three things follow, and together they redirect this stage.

The pushforward on B cannot discriminate. A flat distribution has no mass to concentrate in the tens or near a thousand, so a question posed as "where does the prior put B?" is answered analytically and tells us nothing about the fit.

The count is the likelihood's choice. The prior has no preference; the heuristic returned 846 and the consensus 1,268 because the marginal likelihood selected them. A fine partition therefore cannot be read as the prior's doing, and nothing about the base count was settled before the citations were read.

The expensive assumption is elsewhere. What makes a fine partition cheap is the nested prior over the count matrices, equation (8): a $B \times B$ matrix need not be described against a flat background, only against the coarser grouping above it. That is a prior over *parameters*, and it acts on $\boldsymbol{b}$ only through the marginal likelihood. A pushforward on B is the wrong instrument for it. The right one is the prior predictive of §8, and specifically its flat-against-nested arm.

7.2 What the pushforward is still good for, and why it needs no simulation

Two quantities are not uniform and are worth having. Both come from $P(\boldsymbol{b})$ alone – neither touches $\boldsymbol{e}$ or E – and both have closed forms, so Stage one is arithmetic rather than sampling. The figures below were computed by the recursion in Algorithm 1 and should be reproduced before they are relied on.

The share of singleton groups. $P(\boldsymbol{n} \mid B)$ is uniform over compositions of N into B positive parts, and under that law the probability that a given group has exactly one member is

$$P(n_r = 1 \mid B) \;=\; \frac{\binom{N-2}{B-2}}{\binom{N-1}{B-1}} \;=\; \frac{B-1}{N-1}, \tag{9}$$

which by exchangeability is also the expected share of singleton groups. Conditional on $B = 1{,}268$ the prior expects 35.6 percent singletons; conditional on $B = 846$, 23.8 percent. The fit's 54 percent exceeds both. The marginal figure, averaged over the uniform B, is exactly 50 percent, which is close to the fit – but that closeness is an artifact of the uniform B, half of whose mass sits above 1,779 groups where nearly every group is a singleton. The conditional figures are the ones that bear on the fit, and they say the prior did not anticipate this many singletons.

Hierarchy depth. Write d for the number of levels until the chain $B_0 = N,\ B_l \sim \text{Uniform}\{1, \ldots, B_{l-1}\}$ first reaches one group. A level may repeat the count above it, since the uniform draw includes B_{l-1} itself, and such a level still counts toward the depth. The distribution of d is a one-dimensional recursion over B, exact and instantaneous. At $N = 3{,}557$ the mean depth is 9.75 levels, and the probability of eleven or more is 0.38. Our fit descends through ten non-degenerate levels to a single group at the eleventh, so a hierarchy of this depth is ordinary under the prior rather than extreme. Conditioning on the fitted base count moves the mean to 9.72 given $B_1 = 1{,}268$ and to 9.32 given $B_1 = 846$; neither makes eleven levels remarkable.

Some level in the elicited range. This one needs no recursion. For a band $[\text{lo}, \text{hi}]$ and any starting count above it, the probability that the chain passes through a level inside the band is

$$P\big(\text{some } B_\ell \in [\text{lo}, \text{hi}]\big) \;=\; \frac{\text{hi} - \text{lo} + 1}{\text{hi}}, \tag{10}$$

exactly, independent of N and of B_1. The identity follows from self-consistency: write $v(b)$ for the probability of reaching the band from $b > \text{hi}$; the next count is uniform on $\{1, \dots, b\}$, so $v(b) = \frac{1}{b}\big[v(b) + \sum_{b'<b} v(b')\big]$, and the constant $v = (\text{hi} - \text{lo} + 1)/\text{hi}$ solves this at every b above the band once it holds at $b = \text{hi} + 1$, where it is immediate. For twenty to thirty groups the figure is $11/30 = 0.367$; for the wider band of ten to forty it is $31/40 = 0.775$. The prior gives a readable level with probability a little over one in three, whatever the fit did, which is neither the near-certainty the domain expectation would want nor a rejection of it.

7.3 The domain threshold

The threshold cannot be stated as "$P(B_\ell > 100) \approx 0.01$ at the cut intended for ℓ," because which cut becomes ℓ is the output of this whole process; a threshold defined by reference to its own result is circular. Two non-circular forms are available. Fix the level in advance – declare that ℓ will be read from level five, say – and the threshold applies to a well-defined quantity. Or state it as a property of the hierarchy: the prior should place substantial mass on hierarchies possessing *some* level with between twenty and thirty groups. The second is closer to what the domain knowledge actually supports, since the co-citation literature speaks to how many specialties a field has and not to the depth at which a blockmodel should find them; and it is the form that equation (10) answers in closed form.

7.4 What `graph-tool` does and does not expose

The priors in equations (5) to (7) are written down, so they can be sampled directly in a dozen lines of either language. What cannot be replaced from outside the library is the *fitting*: the samplers and the heuristic are built against those priors internally, and there is no argument for supplying a different $P(\boldsymbol{b})$.

Two partial exceptions should be tried before concluding that nothing is available. `B_min` and `B_max` constrain the number of base groups directly. `clabel` and `pclabel` constrain which vertices may share a group. Both are crude next to a stated prior – a hard interval is not a distribution – and `B_max` has a further defect: it constrains the *base*, while the domain expectation is about some upper level. A model can hold a thousand base blocks and still deliver twenty-five literatures five levels up, so capping the base at thirty asks for something the expectation never demanded. There is no argument for capping an upper level. `B_max` is therefore used here as a proxy that is known to be too strong, and the reading of its result is set accordingly: the quantity recorded is the nested model's own description length under the cap against its unconstrained value, and a penalty on the scale of the near-free merge in Table 2 – a few percent – would count as acceptable. A penalty on the scale of that table's coarser cuts, tens of percent, would not, but because the cap is too strong a large penalty does not by itself show that the prior is unrevisable; it shows that this argument cannot do the revising.

On the figures above, the first row is the one the arithmetic already selects: 54 percent singletons against an expected 36, and a depth that the prior finds unremarkable. The third row remains open until the `B_max` rerun has been priced. A workflow exists to make assumptions visible and

Algorithm 1 Prior pushforward. Every quantity here follows from equation (5) and the level recursion of equation (8); nothing is sampled and E does not enter.

Input: $N = 3{,}557$; the fitted B_1 (846 or 1,268) and depth (eleven levels to one group); the elicited band $[20, 30]$

Number of groups

$B_1 \sim \text{Uniform}\{1, \ldots, N\}$, mean 1,779 ▷ report; it carries no information about the fit

Singleton share, by equation (9)

$z(B) \leftarrow (B-1)/(N-1)$ for $B \in \{846,\ 1{,}268\}$ ▷ 35.6 and 23.8 percent; marginal over B is 50 percent

Depth, by recursion over B

$p_0 \leftarrow \delta_N$ ▷ or δ_{B_1} to condition on the fit

for $l = 1, 2, \ldots$ until the mass at $B = 1$ exceeds $1 - 10^{-6}$ **do**

  $p_l(b) \leftarrow \sum_{b' \geq b} p_{l-1}(b')/b'$ for $b \geq 2$; mass newly arriving at $b = 1$ is $P(d = l)$ ▷ the sum includes $b' = b$: a level may repeat the count above it

end for

A level in the elicited band, by equation (10)

$P(\text{some } B_l \in [20, 30]) \leftarrow 11/30$ ▷ exact; the same for every N and B_1 above the band

Output: $P(d)$ and its mean; $P(d \geq 11)$; $z(B_1)$; $11/30$

Decisions: is 54 percent singletons above $z(B_1)$? is $P(d \geq 11)$ small? is $11/30$ an acceptable chance of a readable level?

Then: rerun the heuristic under `B_max` $= 30$ and record the nested description length against the unconstrained fit, reading the result as §7 directs

Table 3: Outcomes of Stage one and what each requires. The rows concern depth and group sizes; the number of groups is uniform under the prior and is not at issue here.

If the prior	Then	And we
expects fewer singletons than the fit returned, and depths like the fit's	the singletons are the data or the likelihood talking; the depth is not evidence of anything	carry the singleton question to Stage two, where the nested prior over $\boldsymbol{e}$ can be isolated
expects at least as many singletons as the fit returned, conditional on B_1	that part of the fit's shape was anticipated by the prior, independent of any citation	report it, and stop treating the base-level group sizes as a finding about the field
gives a level in the elicited band with probability $11/30$, which the domain expectation finds too low, and `B_max` cannot reach the band without a penalty of tens of percent	the available arguments cannot bring the model into line with domain knowledge	reimplement the degree-corrected blockmodel with a hand-written sampler and a stated prior

Algorithm 2 Prior predictive check, simulating the hierarchy as fitted. Nothing here fits a model, so it needs no library: `generate_sbm` is a convenience, and the statistics are language-neutral.

Input: N; M draws; observed statistics $\boldsymbol{T}_{\text{obs}}$; E held at its observed value

1: **for** $m = 1$ to M **do**
2: $B \sim P(B)$; $\boldsymbol{n} \sim P(\boldsymbol{n} \mid B)$; $\boldsymbol{b}_{1,m} \sim P(\boldsymbol{b} \mid \boldsymbol{n})$ ▷ equation (5)
3: $\boldsymbol{e}_{1,m} \sim P(\boldsymbol{e} \mid B, E)$, then recurse up the levels of (8) until one group remains ▷ the whole hierarchy, not a base partition with a flat prior bolted on
4: $\boldsymbol{k}_m \sim P(\boldsymbol{k} \mid \boldsymbol{e}_{1,m}, \boldsymbol{b}_{1,m})$
5: $\boldsymbol{A}_m \leftarrow \textsc{GenerateSbm}(\boldsymbol{b}_{1,m}, \boldsymbol{e}_{1,m}, \boldsymbol{k}_m)$, directed
6: $\boldsymbol{T}_m \leftarrow$ the statistics of Table 4
7: **end for**
8: **for** each statistic t **do**
9: report the quantile of t_{obs} within $\{t_m\}$
10: **end for**

Output: containment, not agreement: each t_{obs} should sit inside the bulk of its prior predictive distribution

11: **Separate comparison:** repeating steps 2–7 with a flat $P(\boldsymbol{e})$ in place of the recursion in step 3 isolates what the nested prior contributes. Run it as its own arm, labeled as such, rather than mixing the two.

revisable; if the only revisions available are a hard interval on B and a constraint on which vertices may share a group, then the prior is not revisable in the sense the workflow requires, and demonstrating that is more useful than suspecting it.

8 Stage two: prior predictive checks

A predictive check pushes the prior all the way through the data-generating process and asks whether the networks it produces could plausibly be citation networks. Where the pushforward asks whether the implied shape of the hierarchy is sane, the predictive check asks whether the implied *universe of networks* contains anything resembling ours. This is also the first stage at which E matters, since the count matrices are drawn here and not before.

Procedure. Draw a partition from the prior, draw the between-group counts and the degrees from theirs, generate a network, and compute its summary statistics. Repeat a few hundred times. Then overlay the observed values. The draws are *ancestral*: each quantity is drawn in the order the prior generates it, conditional on the one before.

What to compare, and against what. The statistics worth checking are those the model does not fit by construction. Degree distribution is a poor check, because degree correction reproduces it by design. Table 4 lists the useful ones with their sources.

The interpretation. The check is one of containment, not of agreement. The prior does not have to generate our network; it has to generate a universe of networks that could plausibly include it. If the observed k-core profile sits far in the tail of the prior predictive distribution, the posterior has

Table 4: Statistics for the prior predictive comparison. Forward-in-time citation is not among them: 37 of the 1,098,984 arcs between analytic-era works run forwards, but that is a property of the article network, and the community model has no time in it to reproduce.

Statistic	Observed in our network	Domain expectation and source
Reciprocal community pairs	to be counted: pairs (r, s) with $e_{rs} > 0$ and $e_{sr} > 0$, out of the 36,105 pairs in the fitted graph	Low but not zero. Communities in adjacent eras can cite each other in both directions even though individual works cannot. Distinct from the article-level forward-citation count noted in the caption.
In-degree tail	degree 2 to 522, median 5	Heavy-tailed, from cumulative advantage – Price's account of why well-cited papers attract further citation.
Weight skew	pair weights 1 to 4,646	Community-pair intensity inherits the same skew. A prior implying homogeneous weights is at odds with the data by three orders of magnitude.
k-core profile	coreness 2 to 74, with 125 communities at 74 and no vertex below 2	A dense core with a connected periphery. This is the one statistic that nothing in the model fits, which makes it the sharpest check available.
Group-size distribution	54 percent of fitted groups hold one community	Specialty sizes are skewed but not dominated by singletons. A prior that expects half its groups to be singletons expects something unlike a field.

been straining against the prior from the start – and the 690 hub singletons become a mechanism rather than a mystery.

9 Stage three: posterior checks

Two kinds of checking belong here, and they answer different questions. The first asks whether the machinery works at all. The second asks whether the fitted model can reproduce the world.

9.1 Three questions, not one

Three distinct things went wrong or went unverified, and one procedure will not settle all of them.

Is the inference calibrated? Simulate parameters from the prior, generate a network, fit, and record where the true value falls among the posterior draws. Repeat. If prior, likelihood, and sampler are all correct, those ranks are uniform. The summary statistic here is the number of

Algorithm 3 Simulation-based calibration. Answers the first question only; reproducibility and consensus have their own checks. Language follows whichever implementation is being validated.

Input: M replications; L posterior draws each; a scalar summary f, here the number of groups

for $m = 1$ to M **do**
  $\boldsymbol{b}^* \sim P(\boldsymbol{b})$ ▷ the prior, not a convenient partition
  $\boldsymbol{e}^*, \boldsymbol{k}^* \leftarrow$ draws from their priors, as in Algorithm 2
  $\boldsymbol{A} \leftarrow \textsc{GenerateSbm}(\boldsymbol{b}^*, \boldsymbol{e}^*, \boldsymbol{k}^*)$
  $\{\boldsymbol{b}^{(1)}, \ldots, \boldsymbol{b}^{(L)}\} \leftarrow$ posterior draws from fitting $\boldsymbol{A}$
  $a \leftarrow \#\{\, l : f(\boldsymbol{b}^{(l)}) < f(\boldsymbol{b}^*)\,\}$; $t \leftarrow \#\{\, l : f(\boldsymbol{b}^{(l)}) = f(\boldsymbol{b}^*)\,\}$
  $r_m \leftarrow a + \textsc{Binomial}(t, 1/2)$ ▷ f is discrete; ties broken at random, or the ranks are lumpy for the wrong reason
end for

Output: $\{r_m\}$ uniform on $\{0, \ldots, L\}$ if prior, likelihood, and sampler are correct

Reading: ranks bunched in the middle mean the posterior is too wide; ranks at the extremes mean it is too narrow; a consistent skew means a bias

Scale: at $N = 3{,}557$ a full fit is 15–20 hours, so $M = 100$ is months. Run at reduced N – a few hundred vertices – with a short chain of L draws at each replication, and say so when reporting. The heuristic alone yields one partition and no ranks

groups, which is discrete, so ties must be broken at random; otherwise the ranks come out lumpy for reasons that have nothing to do with the model.

The planted-partition test already run is not this. Fitting a planted four-block partition of 300 vertices recovered it at an adjusted Rand index of 0.96, which established that the code calls the right functions. It exercised one favorable point in parameter space and says nothing about calibration.

Two practical constraints shape how this can be run. Under the actual prior, B is uniform on $\{1, \ldots, N\}$, so most simulated networks will have around 1,800 groups of two nodes each – far from the regime we care about, and a poor use of compute. And a single fit at full size takes 15–20 hours, which puts $M = 100$ replications in the range of months. Calibration should therefore be run at reduced N – a few hundred vertices – with a short chain at each replication. The chain is not optional: the ranks in Algorithm 3 need L posterior draws, and the heuristic returns a single partition. A check that uses the heuristic alone is a recovery or bias check, not calibration, and should be named as such if it is what gets run. What calibration at reduced N validates is the inference at that scale, which is weaker than validating it at ours and is the available option.

Is the fit reproducible? Two runs on identical input with the same seed returned hierarchies 2,451 nats apart. Calibration will not explain that, because it averages over replications rather than isolating a divergence. Nor is it a mystery. `graph-tool`'s sweeps are parallelized with OpenMP, and parallel reductions do not sum in a fixed order, so non-determinism under more than one thread is documented behavior rather than a fault. The check is a controlled repeat, and its first step is therefore `openmp_set_num_threads(1)`: pin the threads, fix the seed and `PYTHONHASHSEED`, run the heuristic twice in one process, and record the partition after each internal step until the two runs differ. If they no longer differ, the cause was the thread count and the fix is to pin it for every reported fit. If they still differ, the remaining candidates are an unseeded stream and iteration order over an unordered container, and the first point of divergence names which.

Does the consensus recover what is there? The step that reconciles 250 draws into one partition is the least examined part of the pipeline and the most expensive. Algorithm 3 never touches it, since it compares draws against the truth directly. Check it on its own terms: simulate a network from a known partition, fit it, run the consensus, and measure how well the consensus recovers *co-membership* – for each pair of nodes, whether the consensus places them together when the truth does. Co-membership rather than labels, because labels are arbitrary and the question is whether the structure survives the reconciliation.

9.2 Posterior predictive checks

Generate networks from the fitted model and compare them to the observed network on the statistics of Table 4. The question is now whether the model, having seen the data, can reproduce features of it that it was not fitted to. The k-core spike at coreness 74 is again the sharpest test.

9.3 External validation

Journals give an independent partition of the same communities, derived from publication venue rather than from citation, and comparing the two answers the second half of RQ7 of the specification outline, which asks whether the field's structure follows disciplinary affiliation or cuts across it. Agreement is reported with the adjusted Rand index, computed on modal assignments for the headline figure and across posterior draws for the accompanying uncertainty. Two cautions belong in the reporting: the index is sensitive to the number of groups, so the journal partition should be coarsened to comparable granularity first; and a low value is a substantive finding about how the field is organized, not a failure of the method.

This stage is substantive rather than diagnostic, and it is the only one of the three that produces a result for the paper.

10 Order of operations

1. **Settle the likelihood.** Fit twice, once with counts as multiplicities and once with counts as an edge covariate, and compare the resulting group counts and description-length curves. Everything downstream depends on which is in place, so nothing else can be checked first.

2. **Pushforward on depth and group sizes.** The marginal over the number of groups is uniform by equation (5); the expected singleton share is equation (9); depth and the chance of a level in the elicited band are a short recursion (Algorithm 1). None of it is simulated and none of it depends on E. Then rerun the heuristic under `B_max` $= 30$ and record the nested description length against the unconstrained fit.

3. **Decide the instrument.** If the prior cannot be brought into line with domain knowledge using the arguments `graph-tool` exposes – `B_min`, `B_max`, `clabel`, `pclabel` – the blockmodel moves to a hand-written sampler with a stated prior. Make this decision explicitly and write down the reason.

4. **Prior predictive.** Generate networks from the prior; compare the statistics of Table 4 for containment.

Algorithm 4 Settling the likelihood. Step 1 of the order above, and a prerequisite for Algorithm 2, which is the first step in which E enters.

Input: the community network G with citation counts $\boldsymbol{w}$

1: **for** $\tau \in \{$`eweight`, `recs:discrete-poisson`$\}$ **do**
2: $\quad s_\tau \leftarrow$ MinimizeNestedBlockmodelDl(G, τ)
3: $\quad$ record groups per level, Σ, and E under τ
4: $\quad C_\tau \leftarrow$ the flat description-length curve of Table 2, recomputed under τ
5: **end for**

Output: the two curves, side by side

6: **Reading:** E differs by a factor of sixteen between the two and enters $P(\boldsymbol{e})$ directly, so a large shift in the group count is expected rather than surprising
7: **Note:** this is diagnostic, not decisive. No curve can say whether 4,646 citations between two communities are 4,646 exchangeable events or one intense relationship; that is a modeling judgment, argued in prose and recorded. The comparison shows what the judgment costs
8: **Note:** uses the 45-second heuristic, which sat 18,329 nats above the mode on the real fit. Adequate for a gross comparison of group counts, not for anything finer

5. **Calibration, reproducibility, consensus.** Three separate checks, set out in §9. Simulation-based calibration tests the inference; a controlled repeat locates the non-determinism; a co-membership recovery test assesses the consensus step.

6. **Posterior predictive, then external validation.** The first checks the model against the data; the second checks the literatures against the journals.

Steps one and two are inexpensive: a heuristic fit takes forty-five seconds, and the Stage one quantities are arithmetic. Steps four through six are the substantial work, and there is no sense in starting them before step three has settled which model they apply to.

10.1 Where each step is implemented

The work divides cleanly, and the dividing line is whether a step interrogates `graph-tool`'s assumptions or replaces them.

The priors are written down in equations (5) to (7), so nothing that only reads or samples them needs the library. Algorithm 1 is arithmetic, and Algorithm 2 is a dozen lines of ancestral sampling in either language. What does require Python is anything that *fits*: Algorithm 4, the sampler inside Algorithm 3, and the `B_max` rerun of step 2 all call `graph-tool`'s heuristic or its chain, and those are built against the priors internally. The constraint is on the fitting, not on the prior.

Anything we build ourselves should be Julia, and the usual argument for Python evaporates at exactly the point where we stop using the package. Three reasons, in order of force.

No probabilistic programming language handles this inference well. A partition is a discrete object with $B!$ label symmetry. Hamiltonian Monte Carlo cannot move it, and Stan cannot represent discrete parameters at all. Turing's composable Gibbs and NumPyro's `DiscreteHMCGibbs` can represent them, so the door is not closed – but single-site updates on 3,557 discrete labels mix badly, the label symmetry still bites, and neither offers the merge-and-split proposals that make this tractable. The realistic sampler is a bespoke one: collapsed Gibbs over node reassignments, or merge-and-split moves of the kind `graph-tool` uses. So the comparison is between two languages for hand-written Markov chain Monte Carlo, not between two PPLs.

That sampler is a tight loop. Each sweep touches 3,557 nodes and 36,105 edges, with a bookkeeping update at every reassignment. Julia runs such loops at compiled speed as written. Python requires `numba`, `cython`, or a C extension to approach it, which means the inner loop leaves Python anyway.

The project already has a Julia lane. Julia 1.11.9 is in the container, the era sweeps run there, and the Pajek readers and era-resolution code sit beside them.

So Python's advantage here was always that someone had already written the model. Step 3 of the order above is precisely the decision to stop relying on that, and once it is taken the advantage is gone.

11 What is at stake downstream

Model I cannot be fitted until ℓ exists, so the blockmodel sits on the critical path. The shape of what it returns also determines what the model can say, and two questions about that shape are still open.

How ℓ attaches to a pair. Every community carries a literature label, but the slope $\alpha_{S,\ell}$ multiplies S_{ijp}, which belongs to a *pair*. Communities i and j will often sit in different literatures, and the specification does not say what happens then. Three readings are available and they are not equivalent: index the slope by the citing community's literature, so $\ell = \ell_i$; index it by whether the pair is within or across literatures, giving a two-level contrast; or carry an effect for each of the two literatures and let them add. The first is the narrowest reading of the Model I table in §13.2 of the specification outline, which lists `literature_i` and `literature_j` as separate columns without saying how the slope consumes them. This needs settling before any cut is chosen, because the three readings make different demands on how many literatures there can usefully be.

Whether a fine cut can be estimated at all. A grouping into 1,268 literatures is not collinear with the community effects – θ_i^K and ψ_j^K are random intercepts, while $\alpha_{S,\ell}$ is a random slope on topical proximity, and those are different objects. The real obstacle is replication. Estimating a slope within a literature needs pairs assigned to it, and needs those pairs to vary in S_{ijp}; with 2.8 communities per group at the base, most groups would contribute a handful of pairs spanning a narrow range of proximity, and σ_ℓ would be estimated almost entirely from the prior. The quantity to report before choosing a cut is therefore simple and has not yet been computed: for each candidate cut, the distribution across literatures of the number of pairs and the spread of S_{ijp} within them.

Where the candidates come from. Four numbers have been proposed and they have different standing. **405** is the lowest-scoring cut in Table 2 and is a statement about description length alone; nobody has argued it is interpretable, and at 8.8 communities per group it faces the replication problem above in milder form. **168** is the coarsest cut still within 2 percent of that score. **25** is the cut at which, by the audit of §4, assignment confidence stops improving, 24 of 25 groups span more than one era, and no group is a singleton. **8** is a reading of the hierarchy's shape; the audit prices it at 123 percent above the best cut and finds two of its eight groups holding 2 percent of the communities between them.

The distribution of pairs across literatures, and of S_{ijp} within them, is the criterion that would separate these on grounds the model cares about, and it is more informative than any of the arguments rehearsed above.

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